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Table 1: PPO

1	$\pi_\theta(a s)$	$V_\phi(s)$	θ	ϕ
2	$N \quad \tau^i = (s_0^i, a_0^i, r_0^i, \dots, s_{T^i}^i) \quad i = 1, \dots, N$ $\hat{A}_t^i$ $K \quad \text{epoch}$ $\tau^i \quad t$ $r_t(\theta) = \frac{\pi_\theta(a_t^i s_t^i)}{\pi_{\theta_{old}}(a_t^i s_t^i)}$ PPO - Clip $L^{CLIP}(\theta) = E_{(s,a) \sim \pi_{\theta_{old}}} [\min(r_t(\theta) \hat{A}_t^i, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t^i)]$ $L^{VF}(\phi) = E_{(s,a) \sim \pi_{\theta_{old}}} [(V_\phi(s) - \hat{V}(s))^2]$ $S(\pi_\theta) = E_{s \sim \pi_{\theta_{old}}} [H(\pi_\theta(\cdot s))]$ $L(\theta, \phi) = -L^{CLIP}(\theta) + c_1 L^{VF}(\phi) - c_2 S(\pi_\theta)$ $\theta \quad \phi$ $\theta \leftarrow \theta - \alpha \nabla_\theta L(\theta, \phi)$ $\phi \leftarrow \phi - \alpha \nabla_\phi L(\theta, \phi)$ $\theta_{old} \leftarrow \theta$			